

Linear Algebra I: HW4 Partial Solution

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If T is a linear transformation between two vector spaces, we denote $\text{im } T := R(T)$, $\ker T := N(T)$.

1. TFFTFFTF

12. No. If T is linear, then $T(-2, 0, -6) = (-2) \cdot T(1, 0, 3) = (-2, -2)$.

13. Consider the equation $a_1v_1 + \cdots + a_nv_n = 0_V$, then

$$a_1w_1 + \cdots + a_nw_n = T(a_1v_1 + \cdots + a_nv_n) = T(0_V) = 0_W,$$

which forces that $a_1 = a_2 = \cdots = a_n = 0$.

14. (a) Suppose T is 1-1, and let S be a linearly independent set, if there is a relation $T(S)$, say

$$a_1T(v_1) + \cdots + a_nT(v_n) = 0_W$$

Then $a_1v_1 + \cdots + a_nv_n = 0_V$ since T is 1-1 and therefore $a_1 = a_2 = \cdots = a_n = 0$ since S is linearly independent. The converse can be proved by contradiction.

(b) Any nontrivial relation among S corresponds to a nontrivial relation among $T(S)$ via T since T is 1-1.

(c) Since β is linear independent, $T(\beta)$ is also linear independent with n element. Furthermore, since T is 1-1 and onto, $\dim V = \dim W = n$, thus $T(\beta)$ is a basis for W .

17. (a) $\dim W > \dim V = \dim \text{im } T + \dim \ker T \geq \dim \text{im } T$.

(b) $\dim \ker T = \dim V - \dim \text{im } T > \dim W - \dim \text{im } T$.

27. (a) Recall that there exists a subspace W' of V such that $V = W \oplus W'$. Therefore, for any $v \in V$, v can be uniquely written as $w + w'$, in this case, define $T(v) = w$, then T is a projection on W along W' .

(b) Let $V = \mathbb{R}^2$, $W = \{(x, 0) \mid x \in \mathbb{R}\}$, $W' = \{(0, y) \mid y \in \mathbb{R}\}$, $W'' = \{(y, y) \mid y \in \mathbb{R}\}$. Then $V = W \oplus W' = W \oplus W''$, and

$$T'((a, b)) = (a, 0)$$

$$T''((a, b)) = (a - b, 0)$$

is the projection on W along W' (resp. along W'').

35. (a) Since $V = \text{im } T + \ker T$,

$$\dim V = \dim \text{im } T + \dim \ker T - \dim(\text{im } T \cap \ker T).$$

On the other hand, we have

$$\dim V = \dim \ker T + \dim \text{im } T$$

Therefore $\dim(\text{im } T \cap \ker T) = 0$, $\text{im } T \cap \ker T = \{0_V\}$.

(b) Since $\text{im } T + \ker T \subseteq V$ and $\dim V = \dim \ker T + \dim \text{im } T$, we have $V = \ker T + \text{im } T$. Together with $\ker T \cap \text{im } T$, the statement follows.

Remark. Finite-dimensionality is used when we have an equation related to dimensions.

2. We first claim that $\text{span } T(S) = T(\text{span}(S))$ for any subset S of V . It is clearly true when $S = \emptyset$ ($\{0\} = \{0\}$). We may assume that S is nonempty, in this case, for any $v \in \text{span}(S)$, v can be written as

$$v = a_1 s_1 + \cdots + a_n s_n \text{ for some } a_1, \dots, a_n \in F, s_1, \dots, s_n \in F.$$

Then

$$T(v) = T(a_1 s_1 + \cdots + a_n s_n) = a_1 T(s_1) + \cdots + a_n T(s_n) \in \text{span } T(S),$$

which proves $T(\text{span}(S)) \subseteq \text{span } T(S)$.

Given $v \in \text{span } T(S)$, v can be written as

$$v = a_1 w_1 + \cdots + a_n w_n \text{ for some } a_1, \dots, a_n \in F, w_1, \dots, w_n \in T(S).$$

For each $i = 1, \dots, n$, there exists $s_i \in S$ such that $T(s_i) = w_i$. Thus, we have

$$\begin{aligned} v &= a_1 T(s_1) + a_2 T(s_2) + \cdots + a_n T(s_n) \\ &= T(a_1 s_1 + \cdots + a_n s_n) \in T(\text{span}(S)), \end{aligned}$$

which proves the other side of the containment, and the proof of claim is done. From the claim, since $\text{span}(S) = V$, $\text{span } T(S) = T(\text{span}(S)) = T(V) = \text{im } T$.